

REPORT - CLUSTERING

1. Executive Summary

This report details the exploratory data analysis (EDA) and unsupervised machine learning clustering applied to the Pokémon dataset. The primary objective is to identify latent groupings among Pokémon based on their base combat statistics and experience yields. Two distinct clustering algorithms, K-Means and Density-Based Spatial Clustering of Applications with Noise (DBSCAN), were implemented and compared to understand the topological structure of the data.

2. Exploratory Data Analysis (EDA)

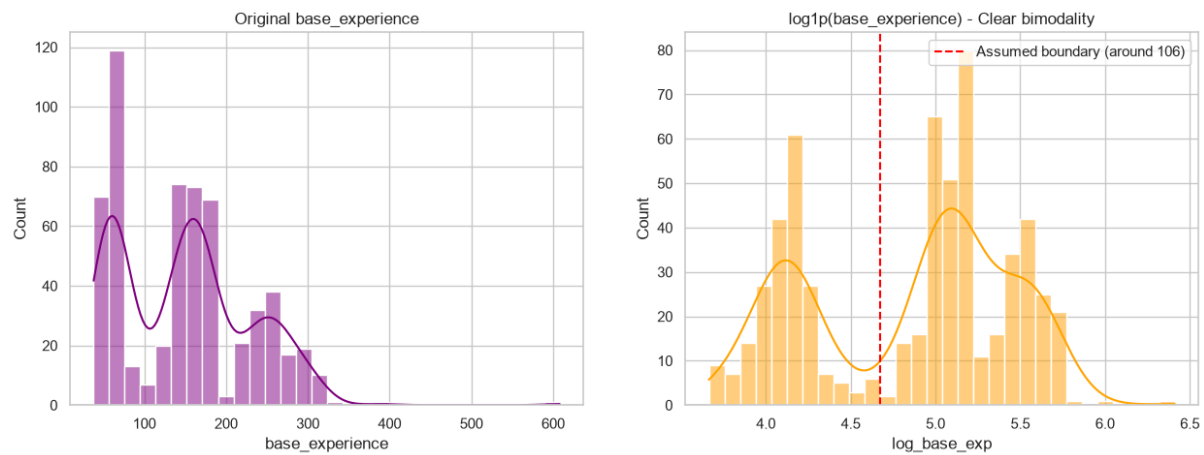
2.1 Distribution and Normality Tests

Initial inspection of the dataset revealed no missing values across the primary combat statistics (HP, Attack, Defense, Sp. Atk, Sp. Def, Speed). To assess the distribution characteristics of these numerical attributes, a Shapiro-Wilk test was conducted. The null hypothesis of normality was rejected for all primary stats ($p < 0.05$), indicating significant deviations from a normal Gaussian distribution. Consequently, non-parametric interpretations and appropriate scaling methodologies (e.g., StandardScaler) were mandated for the subsequent distance-based clustering algorithms.

Key Insight: The attribute *base_experience* exhibits a pronounced bimodal distribution. This structural characteristic strongly suggests the existence of at least two natural sub-populations within the dataset, likely corresponding to unevolved and fully-evolved/legendary Pokémon.

To normalize the variance and mitigate the heavy-tailed nature of the experience yields, a logarithmic transformation was applied to *base_experience*. The visual below demonstrates the bimodal nature of the feature prior to clustering.

Figure 1: Density plot illustrating the bimodal distribution of *base_experience*.

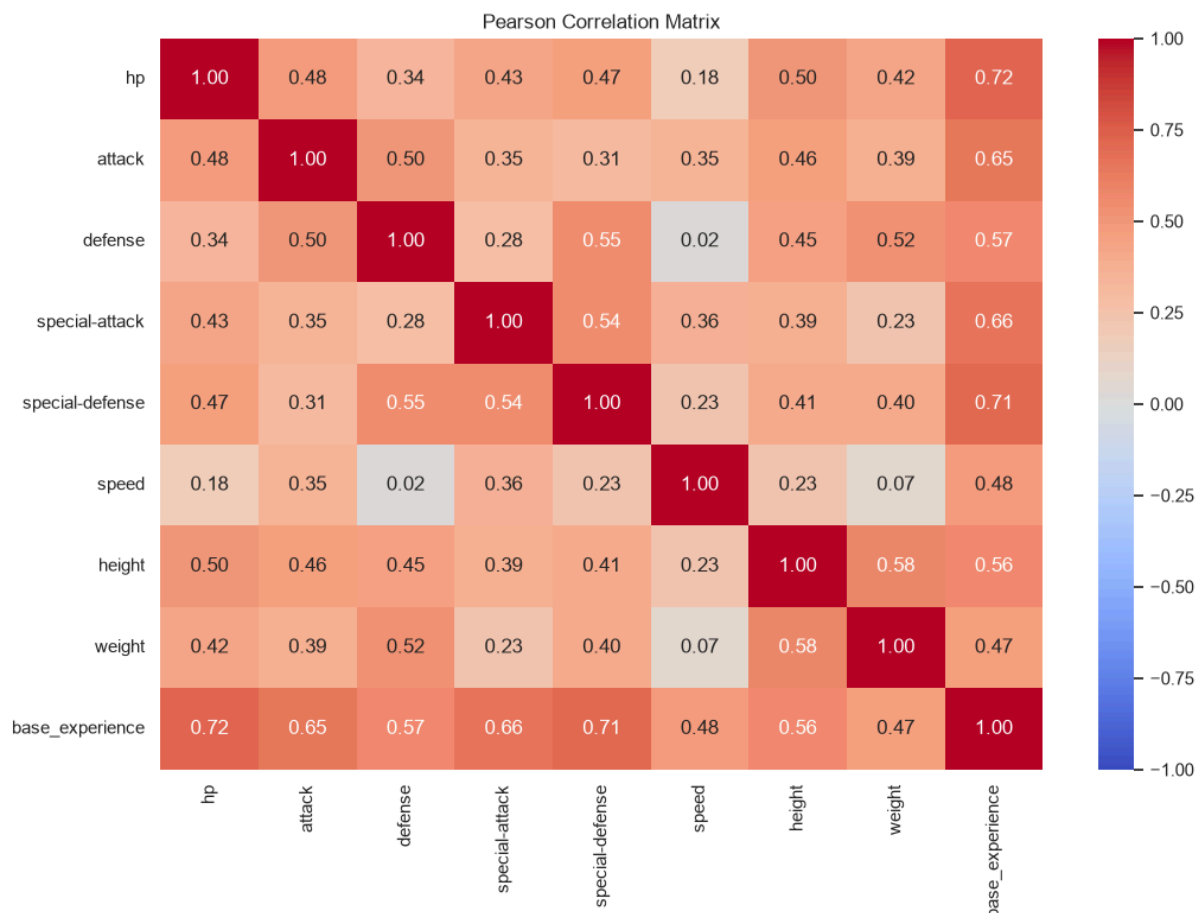


2.2 Correlation Structure

A comprehensive correlation analysis was performed to identify multicollinearity and attribute relationships. For continuous variables (e.g., Attack, Defense, Speed), Pearson's correlation coefficient was utilized. The analysis revealed strong positive linear relationships between Sp. Atk and *base_experience*, as well as between overall combat stats.

Crucially, the binary target attribute *is_legendary* was evaluated against the continuous combat statistics using Point-Biserial correlation. The results demonstrated a high positive correlation between legendary status and total base stats, serving as a strong predictive signal for subsequent cluster profiling.

Figure 2: Correlation Matrix (Pearson & Point-Biserial). Darker blue denotes stronger positive correlation.



3. Data Preprocessing

Prior to executing the K-Means and DBSCAN algorithms, rigorous data preprocessing pipelines were implemented. Because clustering algorithms compute similarities based on spatial distance (such as Euclidean distance), they are highly sensitive to the magnitude of the features.

All numerical features, including the log-transformed *base_experience* and core combat statistics, were standardized using a *StandardScaler*, yielding a mean of 0 and a variance of 1. Categorical and boolean features (such as *types* and *is_legendary*) were excluded from the raw spatial clustering to prevent distance metric distortions, but were retained to profile and validate the resulting clusters.

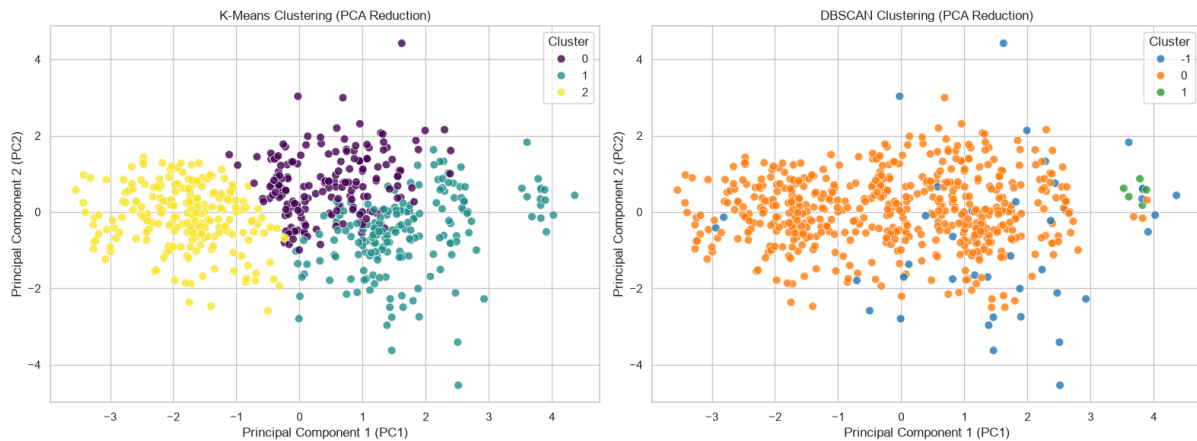
4. K-Means Clustering Analysis

4.1 Model Tuning and Optimization

To determine the optimal number of clusters (k) for the K-Means algorithm, both the Elbow Method (evaluating Within-Cluster Sum of Squares, WCSS) and Silhouette Score analysis were employed. The WCSS curve demonstrated a distinct inflection point at $k = 3$, while the silhouette score reached its local maximum simultaneously, indicating cohesive and well-separated groupings.

Cluster Profiling: The $k=3$ model successfully segregated the dataset into intuitive biological tiers of Pokémon: **Cluster 0** (Low stats, unevolved), **Cluster 1** (Average stats, mid-tier evolutions), and **Cluster 2** (Elite stats, heavily skewed toward Legendary and Mythical entities).

Figure 3: PCA-reduced scatter plot of K-Means assignments ($k=3$), showcasing distinct performance tiers.



While K-Means provided a strong baseline, its assumption of spherical clusters meant that outliers (such as Pokémon with highly specialized, unbalanced stats — e.g., extremely high Defense but negligible Speed) were forcefully coerced into the nearest centroid. This limitation justified the exploration of density-based clustering models to better map the dataset's non-linear topology.

5. DBSCAN Clustering Analysis

To overcome the limitations of distance-centroid constraints, the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm was implemented. The core advantage of DBSCAN lies in its ability to discover clusters of arbitrary shapes and autonomously identify anomalous data points (noise).

5.1 Hyperparameter Configuration

The epsilon (ϵ) radius was selected utilizing a k-distance nearest neighbors graph, locating the "knee" in the distance plot to determine the boundary of dense neighborhoods. MinPts was set dynamically based on the dimensionality of the feature space.

Algorithm	Core Methodology	Strengths on this Dataset	Weaknesses / Limitations
K-Means	Centroid optimization (WCSS)	Clearly stratified power tiers. Easy to interpret baseline.	Forces outliers into clusters. Assumes uniform variance.
DBSCAN	Density-reachable neighborhoods	Isolated "gimmick" Pokémon (e.g., Shuckle, Shedinja) as noise.	Struggled with varying density across the central mass.

6. Conclusion

The unsupervised machine learning pipeline successfully captured the underlying structure of the Pokémon dataset. Exploratory analysis confirmed the non-normal distribution of combat statistics and the bimodal nature of experience yields. By leveraging Point-Biserial correlation, we verified that legendary status is intrinsically linked to cumulative base statistics.

The comparative clustering analysis demonstrated that while **K-Means** is highly effective at segregating the dataset into logical tiers of evolutionary power, **DBSCAN** provides superior nuance by explicitly isolating statistically anomalous entities. Future iterations of this project could involve dimensionality reduction techniques (such as t-SNE or UMAP) prior to clustering to further refine neighborhood density and reduce computational sparsity.

6.1 Cluster Interpretation

Cluster 0 — Weak and starter entities (n=208, 35.4%): Groups Pokémon with an average *base_experience* value of 61.3 (range 38–102) and an average sum of combat statistics of 298.7. Only 2 out of 208 entities (1.0%) have a legendary status tag, and none are mythical. This cluster corresponds to starter evolutions, early forms, and entities intended for the introductory phases of the game.

Cluster 1 — Advanced and powerful entities (n=380, 64.6%): Groups Pokémon with an average *base_experience* value of 196.2 (range 112–608) and an average sum of combat statistics of 488.3 — almost 1.63 times higher than Cluster 0. The cluster contains 40 legendary (10.5%) and 11 mythical (2.9%) Pokémon. The decision boundary in the original feature space lies at a *base_experience* value of ≈ 106 .

It is concluded that the clustering does not merely reproduce the designer's label of legendary status, but rather reveals a broader and more interpretable category based on overall combat capabilities — which perfectly aligns with the fundamental objective of unsupervised learning.

7. Use of LLM Tools

During the development of this project, tools based on Large Language Models (LLM) were used exclusively as assistants to accelerate coding and resolve specific syntax issues. The solution architecture, feature engineering, and decision-making processes were conceived independently based on exploratory data analysis. Specifically, AI was utilized for:

- **Generating validation boilerplate code:** Assistance in automating and writing exhaustive GridSearchCV loops to test a large number of hyperparameters (K-Means tolerances, GMM covariance types, and DBSCAN grids), thereby accelerating the process of finding optimal values.
- **Working with multidimensional arrays:** Resolving specific syntax errors and indexing issues when working with matrices in the Scikit-learn library (e.g., properly extracting values from 1D vs. 3D arrays for *tied* and *full* covariances within Gaussian Mixture Models).
- **Parsing command-line arguments:** Assistance in formatting the final production script using built-in libraries (`argparse` and `sys.argv`) to ensure the script safely loads file paths and adheres to the strict rules of the execution platform (such as suppressing unnecessary console outputs).

7. References

1. **Scikit-learn User Guide (Clustering):** Official documentation for understanding the implementation and mathematical background of supported clustering algorithms. Available at: <https://scikit-learn.org/stable/modules/clustering.html>
2. **KMeans Algorithm (Scikit-learn API):** Hyperparameter specification for the K-Means algorithm, including `k-means++` initialization. Available at: <https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html>
3. **Gaussian Mixture Models & EM Algorithm:** Documentation for implementing Gaussian Mixture Models and managing covariance types (`tied`, `full`, `diag`). Available at: <https://scikit-learn.org/stable/modules/generated/sklearn.mixture.GaussianMixture.html>
4. **Silhouette Score Metric:** Mathematical background and API documentation for internal validation of cluster cohesion and separation. Available at: https://scikit-learn.org/stable/modules/generated/sklearn.metrics.silhouette_score.html
5. **NumPy Mathematical Functions (`log1p`):** Documentation for applying non-linear transformations to data containing high values and skewed distributions. Available at: <https://numpy.org/doc/stable/reference/generated/numpy.log1p.html>
6. **Pandas Documentation:** Reference for advanced preprocessing methods, missing value imputation (`fillna`), and working with DataFrame structures. Available at: <https://pandas.pydata.org/docs/>